Part 5: Final Project & Team Member List

Team Member :

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Topic of The Final Project

RallyRefute: Checking Badminton Pattern Claims Across Annotated Rallies

Objective and Purpose

This project aims to develop a multimedia system for checking factual shot-pattern claims across multiple annotated rallies within a user-selected badminton source video.

Its purpose is to support systematic video review by helping users examine whether an observed pattern holds within the available annotations and locate relevant examples or exceptions. By connecting claims with case counts and video evidence, the system enables users to inspect the basis of each decision. Conclusions remain restricted to the selected annotated data.

Potential Contribution

The potential contribution is a workflow connecting quantified claims, such as “always,” “never,” or a minimum count, with verification across rallies and video evidence. Each decision should be traceable to relevant annotated events, with explicit limits on data coverage. Existing badminton video retrieval work searches for rallies that match a user’s query. This project focuses on checking whether a claim holds across selected annotated rallies and providing video evidence of counterexamples when found.

Proposed Method

Existing annotations are organized into an event index. A small language model converts the claim into a structured query, which is validated and checked programmatically. Lightweight parser fine-tuning may be explored. The architecture and technologies may be modified during implementation if more suitable, reliable, or practical approaches are identified.

Input

The user selects one indexed source video containing multiple annotated rallies and enters a short claim. The initial scope covers three opening shots and the fourth-shot response.
Example: “After serve → net shot → push shot, is the fourth shot always a clear?”

Expected Output

The expected output is a decision—supported within the annotations, refuted, or insufficient data—together with relevant case counts and a replay link when suitable footage is available. For a refuted “always” claim, the system should identify a valid counterexample. A selection without usable relevant cases should return insufficient data.

Candidate Datasets

The primary candidate is Finebadminton-20K. Its locally inspected release contains 2,066 rallies and 20,757 strokes in 70 JSON files. Complete match coverage and game boundaries are not assumed.

ShuttleSet is an alternative if match/game metadata becomes necessary; video access and timestamp alignment would require verification.